

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2023 Course: B. Tech. (Common to all Branches) Semester : I Subject Name & Code: Engineering Mathematics – I (BTBS 101) Max Marks: 60 Date: Duration: 3 Hrs.		
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.		
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Reduce to the Normal form and find the rank of the given matrix. $A = \begin{bmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 10 & 3 & 9 & 7 \\ 16 & 4 & 12 & 15 \end{bmatrix}$	Understand/ CO1	6
B)	Solve the equations: $4x_1 + 2x_2 + x_3 + 3x_4 = 0;$ $6x_1 + 3x_2 + 4x_3 + 7x_4 = 0;$ $2x_1 + x_2 + x_4 = 0.$	Understand/ CO1	6
C)	Verify the Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$ and hence find A^{-1} . Also determine A^8 .	Understand/ CO1	6
Q.2	Solve Any Two of the following:		12
A)	If $r^2 = x^2 + y^2 + z^2$ and $V = r^m$, prove that $V_{xx} + V_{yy} + V_{zz} = m(m+1)r^{m-2}.$	Understand/ CO2	6
B)	If z is a homogeneous function of degree n in x, y , then prove that $x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} = n(n-1)z.$	Understand/ CO2	6
C)	If $u = f(y-z, z-x, x-y)$, show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0.$	Understand/ CO2	6
Q. 3	Solve any Two of the following:		12
A)	Expand $f(x, y) = e^x \cos y$ at $\left(1, \frac{\pi}{4}\right).$	Understand/ CO3	6
B)	Test the function $f(x, y) = x^4 + y^4 - x^2 - y^2 + 1$ for maxima, minima and saddle point.	Understand/ CO3	6
C)	Find the maximum value of $x^m y^n z^p$ when $x + y + z = a.$	Understand/ CO3	6
Q.4	Solve any Two of the following:		12
A)	Evaluate $\int_0^a \frac{x^7 dx}{\sqrt{a^2 - x^2}}.$	Understand/ CO4	6

B)	Trace the curve $y^2 = \frac{x^2(a^2-x^2)}{a^2+x^2}$ (Lemniscate of Bernoulli).	Understand/ CO4	6
C)	Trace the curve $r = a \sin 3\theta$ (3 Leaved Rose).	Understand/ CO4	6
Q. 5	Solve any Two of the following:		12
A)	Evaluate $\int_1^a \int_1^b \frac{dydx}{xy}$	Understand/ CO5	6
B)	Change the order and Evaluate $\int_0^{\pi/2} \int_x^{\pi/2} \frac{\cos y}{y} dx dy$.	Understand/ CO5	6
C)	Evaluate $\int_0^1 \int_{y^2}^1 \int_0^{1-x} x dz dx dy$.	Understand/ CO5	6
	*** End ***		